CS 4953-903/5463 - 906
Spring, 2026

Sushil K. Prasad

Syllabus

(The official syllabus will be the current version on Canvas)

Course Title: Special Topics: High Performance Machine Learning

Prerequisites: Data Structures, Systems, and Programming skills; Undergraduates need to take UG section, else take instructor's permission.

Instructor: Sushil K. Prasad, sushil.prasad@utsa.edu
E-mail is the best way to reach me – put CS 5463/4953 in the subject line.

Class Time and Place: MW 1:30-2:45pm; San Pedro I - 250

Office Hours: MW – 11am-12pm @ SP1-310G; Office hour may be cancelled occasionally due to academic meetings.

TA: Asiful Prodhan (MdAsifulHoque.Prodhan@my.utsa.edu)

Content: This graduate/undergraduate-level topics course integrates high performance computing techniques and study of literature on foundational and cutting-edge parallel and distributed deep learning (DL) research with hands-on experimentations on ParInt Lab DGX and UTSA ARC clusters as well as UIUC Delta GPU supercomputer awarded through NSF NAIRR initiative.

Paper discussions will delve into techniques in model parallelism, distributed training, and inferencing. Students will study the theoretical advances, understand the research landscape and outstanding open research challenges. Topics may include Parallel and distributed techniques (collective operations and all-reduce, matrix multiplication, CUDA and MPI programming), Deep learning (such as DNN, CNN, and Transformers), Deep learning parallel frameworks (e.g., Horovod, TensorFlow, PyTorch, and AxoNN), Concurrency in DNN operators, Concurrency in Deep Neural Networks (data, model, pipelining, and hybrid parallelism), and Distributed Deep Learning training approaches. We will also explore privacy aspects such as federated learning, resource constrained environments such as edge devices, and emerging models such as spiking neural networks.

The overall goals are as follows:

- A general understanding of relevant parallel processing techniques for shared and distributed memory programming using GPUs and clusters
- A general understanding of parallel and distributed techniques for deep learning training and optimization
- To study theoretical advances and understand the research landscape

Format: After coverage of high performance computing basics, MPI and CUDA programming, and demonstrations of various environments for parallel/distributed training, this class would be taught in a seminar style, with focus on 1 / 2 person teams of students jointly reading research papers, updating past slide decks (and preparing new slides) and presenting to the class, and discussing foundational and emerging material. For term project, student will conduct literature survey and writing survey or implementation papers. See sample literature coverage, and sample term project topics appended later.

Attendance: You may be dropped if you have more than two absences. Students are responsible for all the material covered or assigned. No classes March 9-14 (Spring Break). Last class is on May 4 (Mon).

Withdrawals: Drop Deadline to receive a W: March 30

Grading Policy:

	Undergraduates	**Graduates**
Homework and Seminar Presentations	40%	30%
Quizzes, Paper Critiques, and Participation	20%	20%
Term Project and Presentation	40%	50%
Final Project Due	April 27	April 27
Project Presentation	Last few weeks	Last few weeks

Any absence from scheduled seminar presentations must be arranged with the instructor at least two weeks in advance; it remains the responsibility of the absentee to find their replacement. Final grade would be relative to the class performance for each section. To ensure a grade, however, 90 and above will result in a grade of 'A,' 80-89 a 'B,' 70-79 a 'C' and 65-69 a 'D,' and +/- will be employed.

The UG section will have more options for experimental work for their term project and for hands on exercises with parallel and distributed training of neural networks and underlying methods.

Reference Textbooks: "Introduction to Parallel Computing", 2nd Edition, by Ananth Grama, George Karypis, Vipin Kumar, and Anshul Gupta.

- UTSA library provided access to Grama's book in e-format in our subscription databases – Safari (O'Reilly for Higher Education). Introduction to parallel computing - University of Texas at San Antonio (exlibrisgroup.com); Authors slides https://www.cs.purdue.edu/homes/ayg/book/Slides/
- For additional reference, here are two edited books that I have co-authored: https://tcpp.cs.gsu.edu/curriculum/?q=CDER_Book_Project

"Deep Learning," Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016. http://www.deeplearningbook.org

Course Rules: You are fully responsible for all material presented in class or assigned. Assignment due dates are scheduled in advance. A grade of zero will be recorded for late assignments unless prior arrangements are made or for university excused absences (e.g., military service, medical emergency, etc.). Assignments turned in after the due date (with prior arrangements), but within a week from the due date are penalized 20%. Assignments that are more than one week late may earn 0%. *All assignments and projects must be completed to pass the course.*

Cheating: Students are encouraged to discuss assignments in general and ask clarifying questions, but not offer solutions - using another's/third-party solutions or writing solutions/reports for someone else is cheating and a violation of the University's Honor Code. This includes consulting solutions to assignments/test from previous offerings.

Your submissions, including for term projects, must be your own writing. Phrases, figures and tables employed from other sources must be cited appropriately. You may not borrow multiple sentences or paragraphs from other publications.

Use of LLMs is encouraged to improve your writing and occasionally in coding – however its use should be well documented (how, pros, cons), if and when allowed.

University Policies: For common UTSA syllabus information that is part of every syllabus, see the page Common Syllabus Information | UTSA | The University of Texas at San Antonio. It includes the links and the mandatory administrative information concerning disability services, the academic dishonesty policy, and other important issues.

Additional Academic Support: I encourage you to utilize the academic support services available to you through the Tomás Rivera Center (TRC) to assist you with building study skills, etc. These services are available at no additional cost to you. The TRC has several locations at the Main Campus and is also located at the Downtown Campus. For more information, visit the web site at www.utsa.edu/trcss or call (210) 458-4694 on the Main Campus and (210) 458-2838 on the Downtown Campus.

Exclusion Zone: Pursuant to HOP 9.48, Carrying of Concealed Handguns on Campus, my offices are designated exclusion zone. As set out in Section 30.06, Penal Code (trespass by license holder with a concealed handgun), a person licensed to carry a Concealed Handgun under Subchapter H, Chapter 411 Government Code (handgun licensing law), may not enter this property/office with a concealed handgun.
De conformidad con HOP 9.48, Llevar Armas de Fuego Encubiertas en el Campus, mi oficina privada es una zona designada de exclusión. Conforme a la sección 30.06 del código penal (trespasar portando armas de fuego) personas con licencia bajo del sub-capitulo H, capitulo 411, codigo de gobierno (ley de portar armas), no deben entrar a esta propiedad portando un arma de fuego.

Disclaimer: The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus. (Jan 13, 2026)

A teacher can never truly teach unless he is still learning himself. A lamp can never light another lamp unless it continues to burn its own flame. The teacher who has come to the end of his subject, who has no living traffic with his knowledge but merely repeats his lesson to his students, can only load their minds, he cannot quicken them.

Rabindranath Tagore, Indian Poet
Nobel Laureate in Literature, 1913

Syllabus Topic	Category	Paper Title
What and Why of Parallel, Distributed and High Performance Computing		
Overview of Parallel Deep Learning		
Guides on how to use computing platforms - ARC, Delta GPU Cluster		
Parallel Programming Platforms		
OpenMP Shared Memory Programming		
GPU Programming using CUDA		
Accelerating application using CUDA		
Principles of Parallel Algorithm Design		
Basic Communication Operations, Message Passing Programming		
MPI programming for distributed computing		
Demo - Tensorflow; Delta CUDA/DL	Tensorflow; NSF Supercomputers intro	
Demo - Data Parallel - Tensorflow	Data Parallelism	
Demo - model parallelism Tensorflow	Model Parallelism; Concurrency in Networks	
Demo - pipelining parallelism Tensorflow:	Piepline Parallelism; Distributed DL	
Demo - hybrid parallelism Tensorflow	Hybrid Parallelism; ; Distributed DL	
Demo Horovod - Polar Remote Sensing Data	Distributed systems	2018-Horovod fast and easy distributed deep learning in Tensorflow
Demo - AxoNN	Distributed systems	
deep learning	DNN: back propagation, gradient decent	1987- Learning Internal Representations by Error Propagation
deep learning	DNN: back propagation, gradient decent	(L12: pp 19-38, slides) - CNN basics, Examples (L13: 4-36, slides+recording) - CNN Training, Forward and Error Calculations, Backward propagation, Gradient Calculations
deep learning	DNN: Application	1989-Backpropagation applied to handwritten zip code recognition
deep learning	DNN: Application	1998-Gradient-based learning applied to document recognition
deep learning	model consistency, SGD	2011-HOGWILD A Lock-Free Approach to Parallelizing Stochastic Gradient Descent.pdf
deep learning	CNN	2012-ImageNet Classification with Deep Convolutional Neural Networks
DL-frameworks	Tensorflow	2016-TensorFlow A System for Large-Scale ML
	> Tensorflow	2015-tensorflow
concurrency in networks	data parallel	2020-PyTorch Distributed Experiences on Accelerating Data Parallel Training
concurrency in networks	model parallel	2019-Improving Strong-Scaling of CNN Training by Exploiting Finer-Grained Parallelism.pdf

concurrency in networks	pipeline parallel	2019-PipeDream Generalized Pipeline Parallelism for DNN Training
distributed DL	SGD	2012-Large Scale Distributed Deep Networks
distributed DL	parameter distribution: parameter server	2017-Scaling Distributed Machine Learning with the Parameter Server
distributed sytems	distributed, GPUs, python, MPI, NCCL, pytorch	2022-AxoNN An asynchronous, message-driven parallel framework for extreme-scale deep learning
distributed systems	>Horovod, mpi+cuda	2019-Scalable Distributed DNN Training using TensorFlow and CUDA aware MPI characterization design and performance evaluation
distributed systems	hybrid parallel, pytorch, hpc	2022-Hy-Fi Hybrid Five-Dimensional Parallel DNN Training on High Performance GPU Clusters
concurrency in operators	convolution	2006-im2col-High performance convolutional neural networks for document processing
concurrency in operators	convolution	2016-Winograd
deep learning	transformers: LLM	2017-Attention is all you need
concurrency in operators	transformers	2022-Flash attention-Fast and Memory-Efficient Exact Attention with IO-Awareness
deep learning/distributed sytems/LLM	transformers: LLM	2024-Democratizing AI Open-source Scalable LLM Training on GPU based Supercomputers
emerging trends	LLM	2020-Language Models are Few-Shot Learners (GPT3)
emerging trends	SNN	2023-Exploring Neuromotphic Computing based on SNNs
communication	memory	2021-ZeRO-Infinity Breaking the GPU Memory Wall for Extreme Scale Depp Learning.pdf, 2020 - ZeRO Memory Optimizations Toward Training Trillion Parameter Models
communication	>memory, half-precision	2018-mixed_precision_training
emerging trends	neural architecture search	2017-Neural architecture search with reinforcement learning
emerging trends	>edge	2023-Multi-Exit DNN Inference Acceleration Based on Multi-Dimensional Optimization for Edge Intelligence
emerging trends	>edge	2024 -A Converting Autoencoder Toward Low-latency and Energy-efficient DNN Inference at the Edge
emerging trends	federated learning	2017-Communication-Efficient Learning of Deep Networks from Decentralized Data.

Survey-based Term Project

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The goal is to choose a suitable topic, in consultation with the instructor, perform *exhaustive* literature search of the recent material (proceedings, journals, theses/dissertations and technical reports), catalog published material on the chosen topic, browse through the papers and read the important ones, classify the results obtained so far, and write a survey paper on the state-of-the-art the research on the chosen topic. Use ACM Computing Survey papers as guides for style, content, and level of detail (check out samples in the ACM's digital library accessible through UTSA library).

The final survey paper should typically report on (i) 4 foundational papers, (ii) 6 most recent (and relevant) journal papers and (iii) 10 most recent and relevant conference papers to convey the state-of-the-art techniques and knowledge. Other relevant papers are lightly referred to and described.

Course Website: Each student sets up a web site or GitHub for this survey, where all components are to be posted by the following deadlines – ppt/word copies need to be submitted to the instructor and TA as well. For all online references cited, have links available on this site. Use this site to also post and email your presentations a week before your presentation for review. Send a link to your course site within a week.

Deadlines:

- **Topic Selection:** Feb 25
  Submit a half-page write-up defining the scope of your survey. Also, by this date, post the write-up at your course site.

- **Bibliography of literature found:** Mar 4
  This may be a partial list to be complemented as the literature search continues. Have links to these references on your web site.

- **Annotated Bibliography of the literature found:** March 18
  (Scan through the papers and write a brief comment on its content not exceeding 4-5 lines - not from their abstracts/intro. Identify 25 to 30 main papers to be read more thoroughly.)

- **Detailed Annotated Bibliography and Classification of the Results:** April 1
  (i) Detailed comments on papers read thoroughly. (ii) Classification scheme should be a natural one which makes it easier to understand the major developments in an area, and identifies the sub-hierarchies. For each class of major result obtained on the chosen topic, identify the primary and secondary papers associated. The classification scheme will yield the organization of the survey paper.

- **Survey Paper:** Apr 27
  Start with an abstract, explain the topic/problem and define basic terminologies, present various approaches taken or major results obtained justifying your classification, present each major result and briefly discuss minor results within each category giving illustrations where needed, give the current trend and future work remaining in the area, and include the bibliography. All work and explanations must be *adequately referenced* throughout the survey. Have a table of contents in the beginning.

  Appendix contains (1) classification and (2) annotated bibliography.

  A good survey paper can lead to a publication (up to 20 pages + references + appendix).

Term Project : Method Development and Implementation Option

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Choose a suitable problem and dataset, perform a brief literature search, select three recent and competing state-of-the-art parallel/distributed deep learning training algorithms or method optimizations, and implement them on CUDA and/or distributed memory platform (or extend these functionalities in a DL framework). Study their performances (parallel as well as DL-based), perform fine tuning, and write an implementation based paper describing the problem, major approaches, selected algorithms/techniques with justifications, implementations details and performance tuning, performance plots, discussion of plots, and possible future work and your conclusions. A plus would be to come up with your own algorithm (either theoretical or practical).

A good implementation paper can lead to a publication. Employ IEEE/ACM paper format (up to 20 pages + references + appendix).

- Initial half-page writeup containing problem statement: Feb 25
  Link to web site with writeup posted.
- Bibliography and a brief description of your chosen algorithms/technique: March 4
  Have links to all online references on your web site.
- Algorithm/Method 1: March 18
  Post all files - source codes, adequately documented, and timing data files.
- Algorithm/Method 2: April 1
  Meld Algo 1 writeup with Algo 2 writeup and post - to become the Implementation paper.
- Algorithm/Method 3 + Implementation paper: Apr 27
  Meld the three algorithms writeups, organize and polish.

With each algorithm/method, submit (and post) the

1. A 1-2 page description (hard copy) of
   1. your problem,
   2. chosen algorithm/method,
   3. associated data structures and datasets, hyperparameters
   4. underlying communication pattern,
   5. amount of read and write contention and synchronization overheads CUDA/MPI program (or, overheads for distributed program, and in other languages)
   6 parallel time complexity for each implementation with a breakdown of communication/synchronization time and of computation time,
   7. timing and other experiment details (description of what parameters have been varied in what range, and how many runs have used for parallel and DL performance data),
   8. performance of each program with reference to the plots for (a) execution/training time and for (b) speedup as number of processes/GPUs varies, (c) training and accuracy, etc., and
   9. your conclusions containing your interpretation of the performance of these methods, their limitations, and possible future improvements.

Sample Term Project Topics

1. Energy Efficient Machine Learning models for Edge Devices
2. Modern Accelerators for Machine Learning
3. Machine Learning over Polar Satellite Datasets
4. Approximate Similarity Search over polygonal/spatial datasets
5. Spatiotemporal machine learning on emerging architectures
6. Modern Accelerators for Machine Learning
7. Quantum Computing and Communication advances for ML
8. Unlearning in Large Language Models
9. Sparse matrix formats and compression for deep learning.
10. Accelerating locality-sensitive hashing with hardware-aware designs for Unsupervised Learning
11. ML optimization for non-convex datasets
12. Comparative analysis of DL frameworks
13. Hybrid parallelization strategies for extreme-scale models (e.g., GPT-like) architectures).
14. Application of ML for NP Complete Problems
15. Asynchronous vs. synchronous gradient updates in distributed settings.
16. ML optimization under heterogeneous client data distributions
17. Accelerating Spiking Neural Network training on neuromorphic hardware
18. Parallel and ML technique in R-Tree and other hierarchical data structures Construction, Search, and Update
19. Training algorithms for low-power spiking models
20. Designing Spiking Neural Networks from Artificial Neural Networks
21. Meta-learning for optimizing DNN hyperparameters
22. Machine learning techniques for speculative computations such as parallel simulation, optimistic databases, etc.
23. Federated Machine learning techniques.
24. AI/ML dataflow graph scheduling
25. Deformable Convolution
26. Sparse Transformers
27. Fourier domain models
28. Mixture of Experts
29. Tensor parallelism
30. Quantum ML

CS 5463/4953 Student Information

Name:

Telephone numbers where you can be reached:

Email:

Optional – Company working for (Address):

Major and Minor

Where and When Data Structures and Systems course completed:

Where and When Algorithms/Parallel Programming course completed (not required):

Where and When ML/DL course completed (not required):

Your expectations of this course:

Any interesting problem that you would like to investigate for your project?